Energy resources and energy transfers

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1 Paper 1 & 2 Units [1]

Which quantity is matched with its correct unit?

A kinetic energy: watt (W)

B work done: joule (J)

C power: joule (J)

D force: kilogram (kg)

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2 · Part 1 Paper 1 & 2 Energy transfers [1]

The key of a clockwork toy is turned to wind up a spring inside the toy. The toy is then put on the floor and released, and it starts to move.

Which energy store of the toy decreases as the toy starts to move?

A chemical

B thermal

C kinetic

D elastic

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3 · Part 1 Paper 1 & 2 Energy transfers [1]

An electric food mixer is supplied with 2500 J of energy. Of this energy, 750 J is wasted as thermal energy and sound. The rest is useful.

What is the efficiency of the food mixer?

A 70%

B 30%

C 143%

D 0.70%

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4 · Part 1 Paper 1 & 2 Energy transfers [1]

Thermal energy from the Sun reaches the Earth after travelling through space, which is almost a vacuum.

By which method can thermal energy be transferred through a vacuum?

A conduction

B convection

C radiation

D conduction and convection

5 · Part 1 Paper 1 & 2 Work and power [1]

On a farm, a horse pulls a cart a distance of 2.0 km along a straight, level road. The horse pulls with a constant force of 300 N in the direction the cart moves.

What is the work done by the horse on the cart?

A 600 J

B 150 J

C 600 000 J

D 6.7 J

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6 · Part 1 Paper 1 & 2 Work and power [1]

During a game, a tennis ball is thrown straight upwards. Air resistance is small enough to ignore.

Which statement about the ball as it rises from the hand to its highest point is correct?

A The gravitational potential energy gained is equal to the kinetic energy lost.

B The kinetic energy of the ball is greatest at the highest point.

C The total energy of the ball decreases as it rises.

D The ball has no gravitational potential energy at the highest point.

7 · Part 1

Paper 1 & 2 | Work and power | [1]

The motor of a crane does 36 000 J of work lifting a load. It takes 1.5 minutes to do this.

power = work done / time taken

What is the power of the crane's motor?

A 24 000 W

B 54 000 W

C 3 240 000 W

D 400 W

8 · Part 1

Paper 2 only | Energy resources and electricity generation | [1]

In a hydroelectric power station, water stored behind a dam high in the hills flows down through pipes. The moving water turns turbines, which drive generators.

Which sequence describes the main energy transfers?

A thermal store of the water → kinetic store of the turbine → transferred electrically

B gravitational store of the water → kinetic store of the water and turbine → transferred electrically

C chemical store of the water → thermal store of the turbine → transferred electrically

D elastic store of the dam → kinetic store of the water → transferred by heating

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9 · Part 1

Paper 1 & 2 | Energy transfers | [4]

In a science lesson, a student uses a catapult to fire a small rubber ball straight upwards. The student pulls back the elastic band of the catapult and then lets go. The ball rises, stops for an instant at its highest point, and then falls back down.

(a) State the energy store that increases when the student stretches the elastic band. [1]

(b) Describe the energy transfers that take place from the moment the band is released until the ball reaches its highest point. [2]

(c) State the principle of conservation of energy. [1]

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10 · Part 1

Paper 1 & 2 | Energy transfers | [8]

The table shows energy data for two different lamps, each switched on for 10 s.

Lamp	Total energy supplied in 10 s / J	Energy transferred as light in 10 s / J
filament lamp	600	30
LED lamp	80	32

(a) Calculate the efficiency of the filament lamp. [2]

(b) Calculate the energy wasted by the LED lamp in 10 s, and state the energy store that this wasted energy is transferred to. [2]

(c) A Sankey diagram is drawn to scale for the filament lamp. The arrow for the total energy supplied is 12 cm wide. Calculate the width of the arrow for the energy transferred as light. [2]

(d) Explain, using the data in the table, why replacing filament lamps with LED lamps reduces the energy that must be supplied to light a room. [2]

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11 · Part 1

Paper 1 & 2 | Energy transfers | [6]

In a kitchen, a metal pan contains cold soup. The pan is placed on a hot electric hob. Only the base of the pan touches the hob.

(a) State the main method by which thermal energy passes through the metal base of the pan. [1]

(b) Describe how thermal energy is transferred through the metal base by this method. [2]

(c) Explain how all of the soup in the pan becomes hot, even though only the soup at the bottom is in contact with the base of the pan. [3]

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12 · Part 1

Paper 1 & 2 | Energy transfers | [7]

On a sunny day at the coast, the land warms up more quickly than the sea. During the day, a gentle wind called a sea breeze blows from the sea towards the land.

(a) State the method by which energy from the Sun reaches the land. [1]

(b) Explain why the sea breeze blows from the sea towards the land during the day. [4]

(c) At night the land cools down more quickly than the sea, so the land becomes cooler than the sea.

Predict the direction in which the breeze blows at night. Give a reason for your answer. [2]

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13 · Part 1

Paper 1 & 2 | Energy transfers | [7]

In this investigation, a student finds out how the type of surface affects how quickly a metal plate absorbs infrared radiation.

Two metal plates are identical except for their surfaces. Plate X is painted matt black. Plate Y is polished shiny silver. An electric heater is placed between the plates so that each plate is 20 cm from the heater. A thermometer is attached to the back of each plate.

The student switches on the heater and measures the time taken for the temperature of each plate to rise by 10 °C. The table shows the results.

Plate	Surface	Time for temperature to rise by 10 °C / s
X	matt black	45
Y	shiny silver	160

(a) State the independent variable in this investigation. [1]

(b) State two variables that the student should keep the same. [2]

(c) Deduce which surface is the better absorber of infrared radiation. Use the results in the table to justify your answer. [2]

(d) Later, the heater is switched off when both plates are at the same temperature.

Predict which plate cools down faster. Explain your answer. [2]

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14 · Part 1

Paper 1 & 2 | Energy transfers | [6]

In this investigation, a student compares how well different materials conduct thermal energy.

Four rods have the same length and the same thickness. They are made of copper, aluminium, steel and glass. A small drawing pin is stuck to one end of each rod with a blob of wax. The other ends of all four rods are placed in a beaker of hot water at the same time. The student measures the time taken for each pin to fall off as the wax melts.

Material	Time for pin to fall off / s
copper	35
aluminium	60
steel	210
glass	pin had not fallen off after 600 s

(a) State the dependent variable in this investigation. [1]

(b) Identify the material that is the best conductor of thermal energy. Give a reason for your answer. [2]

(c) Explain why the rods must all have the same length and thickness. [2]

(d) Suggest one safety precaution the student should take. [1]

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15 · Part 1

Paper 1 & 2 | Energy transfers | [6]

The owner of a house wants to reduce the energy transferred out of the house by heating in cold weather.

(a) Double glazing uses two panes of glass with a thin layer of trapped air between them.

Explain how double glazing reduces the rate of thermal energy transfer through a window. [2]

(b) Loft insulation is a thick layer of fibreglass, which contains many small pockets of trapped air.

Explain how loft insulation reduces the rate of thermal energy transfer through the roof. [2]

(c) The table shows the cost of installing each method and the money it saves on heating bills each year.

Method	Cost to install / $	Saving on heating bills each year / $
loft insulation	600	200
cavity wall insulation	1500	300
double glazing	4800	160

Determine which method pays back its installation cost in the shortest time. Show your working. [2]

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16 · Part 1

Paper 1 & 2 | Energy transfers | [6]

This question is about a vacuum flask, which is used to keep a drink hot. It has these features:

- a glass container with double walls and a vacuum between the walls
- the surfaces of the walls that face the vacuum are silvered (shiny)
- a plastic stopper in the top
- the glass container sits on insulating supports inside an outer plastic case

(a) Give a reason why thermal energy cannot be transferred across a vacuum by conduction or convection. [1]

(b) Explain how the features of the flask reduce the rate at which the hot drink cools. [5]

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17 · Part 1

Paper 1 & 2 | Work and power | [7]

One student, of mass 55 kg, runs up a flight of stairs. The top of the stairs is 3.6 m vertically above the bottom. The student takes 4.5 s to reach the top.

The gravitational field strength g = 10 N/kg.

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(a) Calculate the gravitational potential energy gained by the student. [2]

(b) State the work done by the student against gravity. [1]

(c) The equation linking power, work done and time taken is:

power = work done / time taken

Calculate the power the student develops against gravity. [2]

(d) Another student of the same mass runs up the same stairs in 3.0 s.

Explain how the power of this student compares with the power of the first student. [2]

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18 · Part 1

Paper 1 & 2 | Work and power | [8]

The total mass of a cyclist and her bicycle is 80 kg. She rides along a level road at a speed of 6.0 m/s.

(a) Calculate the kinetic energy of the cyclist and bicycle. [2]

(b) The cyclist speeds up to 12 m/s.

Calculate the new kinetic energy, and state the factor by which the kinetic energy has increased. [2]

(c) On another day, the cyclist is again moving at 6.0 m/s. She applies the brakes and comes to a stop in a distance of 18 m.

Calculate the average braking force. [3]

(d) State the energy store that most of the kinetic energy is transferred to when the cyclist brakes. [1]

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19 · Part 1

Paper 1 & 2 Work and power [7]

During a competition, a pole vaulter of mass 60 kg sprints along the runway at 9.0 m/s before planting the pole and jumping.

The gravitational field strength g = 10 N/kg.

(a) Show that the kinetic energy of the pole vaulter is about 2400 J. [2]

(b) Assume that all of this kinetic energy is transferred to the gravitational store of the pole vaulter.

Calculate the maximum height that the pole vaulter's centre of gravity could rise. [3]

(c) In practice, the pole vaulter's centre of gravity rises less than the height calculated in (b).

Explain why. [2]

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20 · Part 1

Paper 1 & 2 Work and power [5]

In a warehouse, a worker pushes a heavy crate across a level floor. The worker pushes with a horizontal force of 150 N and the crate moves 12 m at a constant speed.

(a) Calculate the work done by the worker on the crate. [2]

(b) The crate's kinetic energy does not change. State the energy store that the work done by the worker is transferred to. [1]

(c) Later, the worker pushes with a force of 150 N against a large cupboard, but the cupboard does not move.

Explain why the worker does no work on the cupboard. [2]

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21 · Part 1

Paper 1 & 2 Work and power [8]

In this question, a car travels along a straight, level road at a constant speed. The driving force from the engine is 600 N. The car travels 500 m in 20 s.

(a) Calculate the work done by the driving force as the car travels 500 m. [2]

(b) The equation linking power, work done and time taken is:

power = work done / time taken

Calculate the useful power output of the engine. [2]

(c) The engine does work on the car, but the kinetic energy of the car does not increase.

Explain why. [2]

(d) The total power input to the engine from the fuel is 60 000 W.

Calculate the efficiency of the engine. [2]

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22 · Part 1 Paper 1 & 2 Work and power [8]

In an experiment, a box of mass 4.0 kg is released from rest at the top of a ramp. The box slides down the ramp. The top of the ramp is 1.5 m vertically above the bottom, and the length of the ramp is 3.0 m. At the bottom of the ramp, the speed of the box is 4.0 m/s.

The gravitational field strength g = 10 N/kg.

(a) Calculate the gravitational potential energy lost by the box. [2]

(b) Calculate the kinetic energy of the box at the bottom of the ramp. [2]

(c) Calculate the work done against friction as the box slides down the ramp. Explain your reasoning. [2]

(d) Calculate the average frictional force on the box as it slides down the ramp. [2]

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23 · Part 1

Paper 2 only

Energy resources and electricity generation [6]

Electricity can be generated from many different energy resources.

(a) Describe the energy transfers that take place in a power station that burns natural gas, from the fuel to the electricity supplied to the grid. [4]

(b) Describe the energy transfer that takes place in a solar cell. [1]

(c) State the energy store of the fuel that a nuclear power station uses. [1]

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24 · Part 1

Paper 2 only

Energy resources and electricity generation [7]

In one region of a country, there are very hot rocks a few kilometres below the ground. The country plans to build a geothermal power station there. It already has a wind farm on hills near the coast.

(a) Describe how energy is transferred when electricity is generated from geothermal resources. [3]

(b) Describe the energy transfers that take place when a wind turbine generates electricity. [2]

(c) State one advantage of geothermal power compared with wind power. [1]

(d) State one disadvantage of geothermal power. [1]

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25 · Part 1

Paper 2 only

Energy resources and electricity generation [7]

One country plans to close a coal-fired power station. Two replacements are being considered: a nuclear power station, or a large solar farm made of solar cells. Each would supply the same total energy over a year.

(a) State what is meant by a renewable energy resource. [1]

(b) Discuss the advantages and disadvantages of each replacement. [6]

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